Topic: Solving Ax=b

Question: Find the general solution to $A\vec{x} = \vec{b}$.

$$A = \begin{bmatrix} 3 & -6 & 6 \\ -3 & 7 & -9 \\ -6 & 8 & 0 \end{bmatrix} \text{ with } \vec{b} = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$$

Answer choices:

A $$\vec{x} = \begin{bmatrix} \frac{5}{3} \\ 3 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$$

B $$\vec{x} = \begin{bmatrix} \frac{1}{3} \\ 3 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$$

C $$\vec{x} = \begin{bmatrix} \frac{5}{3} \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$$

D $$\vec{x} = \begin{bmatrix} \frac{1}{3} \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$$

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Solution: D

We augment the matrix with $\vec{b} = (b_1, b_2, b_3)$.

$$\left[\begin{array}{ccc|c} 3 & -6 & 6 & b_1 \\ -3 & 7 & -9 & b_2 \\ -6 & 8 & 0 & b_3 \end{array}\right]$$

Now we'll put the matrix into reduced row-echelon form.

$$\left[\begin{array}{ccc|c} 1 & -2 & 2 & \frac{1}{3}b_1 \\ -3 & 7 & -9 & b_2 \\ -6 & 8 & 0 & b_3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 2 & \frac{1}{3}b_1 \\ 0 & 1 & -3 & 3(\frac{1}{3}b_1) + b_2 \\ -6 & 8 & 0 & b_3 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 2 & \frac{1}{3}b_1 \\ 0 & 1 & -3 & b_1 + b_2 \\ 0 & -4 & 12 & 6(\frac{1}{3}b_1) + b_3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 2 & \frac{1}{3}b_1 \\ 0 & 1 & -3 & b_1 + b_2 \\ 0 & 0 & 0 & 4(b_1 + b_2) + 2b_1 + b_3 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -4 & 2(b_1 + b_2) + \frac{1}{3}b_1 \\ 0 & 1 & -3 & b_1 + b_2 \\ 0 & 0 & 0 & 6b_1 + 4b_2 + b_3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -4 & \frac{7}{3}b_1 + 2b_2 \\ 0 & 1 & -3 & b_1 + b_2 \\ 0 & 0 & 0 & 6b_1 + 4b_2 + b_3 \end{array}\right]$$

Using the third row of the augmented matrix, we can verify that $\vec{b} = (1, -1, -2)$ will produce a solution $\vec{x}$.

$$6b_1 + 4b_2 + b_3 = 0$$

$$6(1) + 4(-1) + (-2) = 0$$

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$$6 - 4 - 2 = 0$$

$$0 = 0$$

Now that we know we can get a solution, we'll start by finding the complementary solution, starting with the rref matrix that we found earlier.

$$\left[\begin{array}{ccc|c} 1 & 0 & -4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

This matrix produces a system of two equations,

$$1x_1 + 0x_2 - 4x_3 = 0$$

$$0x_1 + 1x_2 - 3x_3 = 0$$

The system simplifies to

$$x_1 = 4x_3$$

$$x_2 = 3x_3$$

Then the vector set that satisfies the null space is given by

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$$

We can therefore write the complementary solution as

$$\vec{x}_n = c_1 \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$$

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Now we need to find the particular solution that satisfies $A\vec{x}_p = \vec{b}$. We'll plug $\vec{b} = (1, -1, -2)$ into the augmented matrix we built earlier.

$$\begin{bmatrix} 1 & 0 & -4 & | & \frac{7}{3}b_1 + 2b_2 \\ 0 & 1 & -3 & | & b_1 + b_2 \\ 0 & 0 & 0 & | & 6b_1 + 4b_2 + b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -4 & | & \frac{7}{3}(1) + 2(-1) \\ 0 & 1 & -3 & | & 1 - 1 \\ 0 & 0 & 0 & | & 6(1) + 4(-1) - 2 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 1 & 0 & -4 & | & \frac{1}{3} \\ 0 & 1 & -3 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

To find a vector $\vec{x}_p$ that satisfies $A\vec{x}_p = \vec{b}$, we need to choose a value for the free variable, and the easiest value to use is $x_3 = 0$. Using that value, we'll rewrite the matrix as a system of equations.

$$1x_1 + 0x_2 - 4(0) = \frac{1}{3}$$

$$0x_1 + 1x_2 - 3(0) = 0$$

The system becomes

$$x_1 = \frac{1}{3}$$

$$x_2 = 0$$

So the particular solution then is $x_1 = 1/3$, $x_2 = 0$, and $x_3 = 0$, or

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$$\vec{x}_p = \begin{bmatrix} \frac{1}{3} \\ 0 \\ 0 \end{bmatrix}$$

We'll get the general solution by adding the particular and complementary solutions.

$$\vec{x} = \vec{x}_p + \vec{x}_n$$

$$\vec{x} = \begin{bmatrix} \frac{1}{3} \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$$

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Topic: Solving Ax=b

Question: Find the general solution to $A\vec{x} = \vec{b}$.

$$A = \begin{bmatrix} 1 & -1 & 3 & -1 & 3 \\ -1 & 0 & 0 & 2 & 1 \\ 0 & 1 & -3 & -1 & -4 \end{bmatrix} \text{ with } \vec{b} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Answer choices:

A $$\vec{x} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 0 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 4 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

B $$\vec{x} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 3 \\ 1 \\ 4 \end{bmatrix}$$

C $$\vec{x} = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 0 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 4 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

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D $$\vec{x} = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 3 \\ 1 \\ 4 \end{bmatrix}$$

Solution: A

We augment the matrix with $\vec{b} = (b_1, b_2, b_3)$.

$$\left[\begin{array}{ccccc|c} 1 & -1 & 3 & -1 & 3 & b_1 \\ -1 & 0 & 0 & 2 & 1 & b_2 \\ 0 & 1 & -3 & -1 & -4 & b_3 \end{array}\right]$$

Now we'll put the matrix into reduced row-echelon form.

$$\left[\begin{array}{ccccc|c} 1 & -1 & 3 & -1 & 3 & b_1 \\ 0 & -1 & 3 & 1 & 4 & b_2 + b_1 \\ 0 & 1 & -3 & -1 & -4 & b_3 \end{array}\right] \rightarrow \left[\begin{array}{ccccc|c} 1 & -1 & 3 & -1 & 3 & b_1 \\ 0 & 1 & -3 & -1 & -4 & -b_2 - b_1 \\ 0 & 1 & -3 & -1 & -4 & b_3 \end{array}\right]$$

$$\left[\begin{array}{ccccc|c} 1 & -1 & 3 & -1 & 3 & b_1 \\ 0 & 1 & -3 & -1 & -4 & -b_2 - b_1 \\ 0 & 0 & 0 & 0 & 0 & b_3 + b_2 + b_1 \end{array}\right] \rightarrow \left[\begin{array}{ccccc|c} 1 & 0 & 0 & -2 & -1 & -b_2 \\ 0 & 1 & -3 & -1 & -4 & -b_2 - b_1 \\ 0 & 0 & 0 & 0 & 0 & b_3 + b_2 + b_1 \end{array}\right]$$

Using the third row of the augmented matrix, we can verify that $\vec{b} = (1, -2, 1)$ will produce a solution $\vec{x}$.

$$b_3 + b_2 + b_1 = 0$$

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$$1 - 2 + 1 = 0$$

$$0 = 0$$

Now that we know we can get a solution, we'll start by finding the complementary solution, starting with the rref matrix that we found earlier.

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & -2 & -1 & 0 \\ 0 & 1 & -3 & -1 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

This matrix produces a system of two equations,

$$x_1 - 2x_4 - x_5 = 0$$

$$x_2 - 3x_3 - x_4 - 4x_5 = 0$$

The system simplifies to

$$x_1 = 2x_4 + x_5$$

$$x_2 = 3x_3 + x_4 + 4x_5$$

Then the vector set that satisfies the null space is given by

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} 0 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 1 \\ 4 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

We can therefore write the complementary solution as

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$$\vec{x_n} = c_1 \begin{bmatrix} 0 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 4 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Now we need to find the particular solution that satisfies $A\vec{x}_p = \vec{b}$. We'll plug $\vec{b} = (1, -2, 1)$ into the augmented matrix we built earlier.

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -1 & | & -b_2 \\ 0 & 1 & -3 & -1 & -4 & | & -b_2 - b_1 \\ 0 & 0 & 0 & 0 & 0 & | & b_3 + b_2 + b_1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & -2 & -1 & | & -(-2) \\ 0 & 1 & -3 & -1 & -4 & | & -(-2) - 1 \\ 0 & 0 & 0 & 0 & 0 & | & 1 - 2 + 1 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -1 & | & 2 \\ 0 & 1 & -3 & -1 & -4 & | & 1 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

To find a vector $\vec{x}_p$ that satisfies $A\vec{x}_p = \vec{b}$, we need to choose values for the free variables, and the easiest values to use are $x_3 = 0$, $x_4 = 0$, and $x_5 = 0$. Using those values, we'll rewrite the matrix as a system of equations.

$$x_1 - 2(0) - 0 = 2$$

$$x_2 - 3(0) - 0 - 4(0) = 1$$

The system becomes

$$x_1 = 2$$

$$x_2 = 1$$

So the particular solution then is $x_1 = 2$, $x_2 = 1$, $x_3 = 0$, $x_4 = 0$, and $x_5 = 0$, or

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$$\vec{x_p} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

We'll get the general solution by adding the particular and complementary solutions.

$$\vec{x} = \vec{x}_p + \vec{x}_n$$

$$\vec{x} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 0 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 4 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

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Topic: Solving Ax=b

Question: Find the general solution to $A\vec{x} = \vec{b}$.

$$A = \begin{bmatrix} 2 & 3 & 4 & -4 \\ 2 & 3 & 8 & -10 \\ 6 & 9 & 16 & -18 \end{bmatrix} \text{ with } \vec{b} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Answer choices:

A $$\vec{x} = \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{4} \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} \frac{3}{2} \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 0 \\ \frac{3}{2} \end{bmatrix}$$

B $$\vec{x} = \begin{bmatrix} \frac{3}{2} \\ 0 \\ -\frac{1}{2} \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -\frac{3}{2} \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ \frac{3}{2} \\ 1 \end{bmatrix}$$

C $$\vec{x} = \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{4} \\ 0 \end{bmatrix}$$

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D $\quad \vec{x} = \begin{bmatrix} 1 \\ 0 \\ \frac{1}{4} \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -\frac{3}{2} \\ 1 \\ 1 \\ \frac{3}{2} \end{bmatrix}$

Solution: B

We augment the matrix with $\vec{b} = (b_1, b_2, b_3)$.

$$\begin{bmatrix} 2 & 3 & 4 & -4 & | & b_1 \\ 2 & 3 & 8 & -10 & | & b_2 \\ 6 & 9 & 16 & -18 & | & b_3 \end{bmatrix}$$

Now we'll put the matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & \frac{3}{2} & 2 & -2 & | & \frac{1}{2}b_1 \\ 2 & 3 & 8 & -10 & | & b_2 \\ 6 & 9 & 16 & -18 & | & b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{3}{2} & 2 & -2 & | & \frac{1}{2}b_1 \\ 0 & 0 & 4 & -6 & | & -2(\frac{1}{2}b_1) + b_2 \\ 6 & 9 & 16 & -18 & | & b_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{3}{2} & 2 & -2 & | & \frac{1}{2}b_1 \\ 0 & 0 & 4 & -6 & | & -b_1 + b_2 \\ 0 & 0 & 4 & -6 & | & -6(\frac{1}{2}b_1) + b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{3}{2} & 2 & -2 & | & \frac{1}{2}b_1 \\ 0 & 0 & 4 & -6 & | & -b_1 + b_2 \\ 0 & 0 & 0 & 0 & | & -3b_1 + b_3 - (-b_1 + b_2) \end{bmatrix}$$

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$$\begin{bmatrix} 1 & \frac{3}{2} & 2 & -2 & | & \frac{1}{2}b_1 \\ 0 & 0 & 1 & -\frac{3}{2} & | & \frac{1}{4}(-b_1+b_2) \\ 0 & 0 & 0 & 0 & | & -2b_1+b_3-b_2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{3}{2} & 0 & 1 & | & -2\cdot\frac{1}{4}(-b_1+b_2)+\frac{1}{2}b_1 \\ 0 & 0 & 1 & -\frac{3}{2} & | & \frac{1}{4}(-b_1+b_2) \\ 0 & 0 & 0 & 0 & | & -2b_1+b_3-b_2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{3}{2} & 0 & 1 & | & b_1-\frac{1}{2}b_2 \\ 0 & 0 & 1 & -\frac{3}{2} & | & \frac{1}{4}(-b_1+b_2) \\ 0 & 0 & 0 & 0 & | & -2b_1+b_3-b_2 \end{bmatrix}$$

Using the third row of the augmented matrix, we can verify that $\vec{b} = (1,-1,1)$ will produce a solution $\vec{x}$.

$$-2b_1+b_3-b_2=0$$

$$-2(1)+1-(-1)=0$$

$$0=0$$

Now that we know we can get a solution, we'll start by finding the complementary solution, starting with the rref matrix that we found earlier.

$$\begin{bmatrix} 1 & \frac{3}{2} & 0 & 1 & | & 0 \\ 0 & 0 & 1 & -\frac{3}{2} & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

This matrix produces a system of two equations,

$$x_1+\frac{3}{2}x_2+x_4=0$$

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$$x_3 - \frac{3}{2}x_4 = 0$$

The system simplifies to

$$x_1 = -\frac{3}{2}x_2 - x_4$$

$$x_3 = \frac{3}{2}x_4$$

Then the vector set that satisfies the null space is given by

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} -\frac{3}{2} \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ \frac{3}{2} \\ 1 \end{bmatrix}$$

We can therefore write the complementary solution as

$$\vec{x}_n = c_1 \begin{bmatrix} -\frac{3}{2} \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ \frac{3}{2} \\ 1 \end{bmatrix}$$

Now we need to find the particular solution that satisfies $A\vec{x}_p = \vec{b}$. We'll plug $\vec{b} = (1, -1, 1)$ into the augmented matrix we built earlier.

$$\begin{bmatrix} 1 & \frac{3}{2} & 0 & 1 & | & b_1 - \frac{1}{2}b_2 \\ 0 & 0 & 1 & -\frac{3}{2} & | & \frac{1}{4}(-b_1 + b_2) \\ 0 & 0 & 0 & 0 & | & -2b_1 + b_3 - b_2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{3}{2} & 0 & 1 & | & 1 - \frac{1}{2}(-1) \\ 0 & 0 & 1 & -\frac{3}{2} & | & \frac{1}{4}(-1 + (-1)) \\ 0 & 0 & 0 & 0 & | & -2(1) + 1 - (-1) \end{bmatrix} \rightarrow$$

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$$\left[\begin{array}{cccc|c} 1 & \frac{3}{2} & 0 & 1 & \frac{3}{2} \\ 0 & 0 & 1 & -\frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

To find a vector $\vec{x}_p$ that satisfies $A\vec{x}_p = \vec{b}$, we need to choose values for the free variables, and the easiest values to use are $x_2 = 0$ and $x_4 = 0$. Using those values, we'll rewrite the matrix as a system of equations.

$$x_1 + \frac{3}{2}(0) + 0 = \frac{3}{2}$$

$$x_3 - \frac{3}{2}(0) = -\frac{1}{2}$$

The system becomes

$$x_1 = \frac{3}{2}$$

$$x_3 = -\frac{1}{2}$$

So the particular solution then is $x_1 = 3/2$, $x_2 = 0$, $x_3 = -1/2$, and $x_4 = 0$, or

$$\vec{x}_p = \begin{bmatrix} \frac{3}{2} \\ 0 \\ -\frac{1}{2} \\ 0 \end{bmatrix}$$

We'll get the general solution by adding the particular and complementary solutions.

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$$\vec{x} = \vec{x}_p + \vec{x}_n$$

$$\vec{x} = \begin{bmatrix} \frac{3}{2} \\ 0 \\ -\frac{1}{2} \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -\frac{3}{2} \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ \frac{3}{2} \\ 1 \end{bmatrix}$$

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